Table 5: Jigsaw Bias Speech Suppression Scores

Identity group	Google	Jigsaw	OpenAI (scores)	OpenAI (flags)	Llama Guard	Anthropic
men	0.33	0.46	0.56	0.0	0.16	1.91
women	0.28	0.49	0.65	0.0	0.15	1.56
straight	0.93	2.28	2.55	1.83	0.41	0.63
lgbt	1.16	3.6	2.26	3.59	0.89	0.7
non-white	0.42	0.67	0.81	0.0	1.75	0.81
white	0.59	1.29	0.72	0.0	1.15	5.15
christian	2.16	0.37	0.35	0.0	0.27	0.17
non-christian	2.16	0.57	0.39	0.1	0.71	0.85
disability	1.65	2.92	1.8	0.0	0.16	0.89

helps us to understand that explicit identity references are associated with some portion of the identified speech suppression, in addition to further suppression based on the text content.

6 DISCUSSION AND LIMITATIONS

Our focus on identity-based speech suppression by LLM content moderation endpoints represents a novel contribution to the content moderation literature. While most content moderation and AI safety research prioritizes preventing harmful content from passing through filters (false negatives), we argue that examining false positives is also important, particularly in generative AI contexts where creative expression may be impacted.

6.1 Balancing false positives and false negatives

While content moderation endpoints for LLMs are trained for company-specific standards for text toxicity, our TV and movie plots datasets represent normative judgments about content appropriateness derived from established rating systems. While we recognize that companies must establish their own content moderation norms aligned with their values and risk tolerances, we believe it is important to audit how these privately trained content moderation filters might diverge from widespread norms for appropriate content (such as the age ratings we use for the TV and movies dataset). Determining the appropriate balance between company-specific priorities and public expectations of content moderation remains an open question.

We also recognize the legitimate, prosocial reasons companies may "err on the side of caution" by over-filtering text. As described in Section 2.3, generative AI systems have produced content that has harmed and endangered real people. The public focus on harm creates well-justified pressure to prioritize preventing false negatives, even at the cost of increased false positives. Nevertheless, our results suggest that these safety-oriented approaches may have unintended consequences for creative expression, particularly for content related to identity.

6.2 Innovations and limitations

One innovation of this work is the use of "in-the-wild" datasets and associated labels — the TV and movie datasets with corresponding age ratings — that we introduce to assess the impact of automated content moderation on creative content. Our focus on identifying falsely flagged content allows us to make use of this human-created content, providing a benchmark for normative societal expectations in these domains. At the same time, this presents a limitation: the nature of the TV and movies dataset prevents us from attributing sole causality for this over-flagging to the presence of particular identity groups in the text, as there are unaccounted-for differences in the distribution of content across TV and movie synopses containing references to different identity groups. It is exactly these differences we are interested in assessing. We can gain some insight into controlled differences through the Jigsaw Bias dataset, which was artificially constructed by substituting identity words into short phrases. The presence of speech suppression even on this dataset demonstrates that references to identity groups on their own must be at least some source of the over-flagging of identity-related speech.

One site of possible disagreement and limitations of this work stems from the foundational question of what it means for a text instance to be "identity-related." In this work, we use a number of schemes to tag specific datasets by identity: *template-based*, in which the identity is explicitly referenced (e.g., "hug gay"); *individually coded*, in which the text has been manually identified by a person as related to the identity; *text references*, in which we identify a text as related to an identity if it includes a slur, slang, or neutral term related to that identity; and *external associations*, in which we find external information associating TV shows and movies with identity groups. We use multiple tagging methods to mitigate the limitations of any one of these methods, casting a wide net to identify "identity-related speech", and find speech suppression across all tagging methods.

The specific flagging and stereotype patterns we identify in identity-related speech suppression are likely influenced by both the identity tagging schemes and the specific datasets and labels used. For example, the regression results (see Section 5.2.1) indicate that the automated content moderation systems tested are more poorly calibrated in flagging longer content, yet these systems more accurately flag the longer generative AI dataset, likely because it contains content that is more "fluent" and easy to categorize relative to the shorter user-generated comments that make up much of the traditional dataset. When considering the performance of APIs on specific identities, it is similarly hard to disentangle the co-occurrence of identity-related and harmful content. The analyses in Sections 5.3 and 5.4 assess and disentangle some of these effects, and some are mitigated through the focus throughout on *incorrect*

suppression, but there may be additional co-occurrence effects that future work could usefully identify.

6.3 Future work

Achieving both precise identification of unwanted content and identity-based fairness in alignment with *any* standard is an unsolved technical challenge. After all, *human*-facilitated content moderation has well-documented difficulties with efficacy and fairness across identity groups (see Section 2); we should therefore expect that, even with good-faith efforts by companies to address fairness issues, fully automated content moderation filters will struggle with both efficacy and fairness.

We see two possible useful paths forward. First, non-automated content moderation has long employed humans to help moderate content in cases of ambiguous policy application, with concerning job conditions and mental health outcomes [33, 60]. While finding ways to incorporate humans-in-the-loop to screen generated items may therefore be not only impractical for generative AI but also inadvisable, there may be ways to incorporate the desires and opinions of generative AI users via carefully designed preferences or interface interactions. Future work could consider how best to draw on user input as part of this process.

Second, with the identification that identity-related text is flagged based on stereotypes and text associations that are specific to different demographic groups (see Section 5.4), there may be targeted approaches that can be taken to decouple incorrectly flagged identity-related text from notions of undesired content. For example, we saw that OpenAI had some success by using a template-based explicit identity term approach in training to overcome some (but not all) undesired text associations when tested on a similar template-based dataset. Purposeful training to remove such associations and stereotypes from these automated content moderation systems may be effective in increasing both accuracy on these incorrectly flagged instances and fairness in the distributions of such errors.

Finally, we also encourage future work that further examines the impact of identity-related speech suppression in creative contexts, for example by working directly with creators. Such work could also usefully examine how these APIs are built on by other apps or software layers, and how these choices impact the overall user experience and resulting creative work.

7 CONCLUSION

Our research shifts focus from the traditional concern of content moderation — filtering out toxic or undesired content — to examine the rate and patterns of false positives: non-toxic content incorrectly flagged by moderation systems. While it is crucial to filter harmful content, the incorrect suppression of legitimate speech carries significant consequences, particularly when unevenly distributed across identity groups. We introduce *speech suppression scores* to quantify this disparity, measuring the ratio between the false positive rate for text referencing particular identity groups versus the overall false positive rate.

To enable a comprehensive analysis of speech suppression in contexts relevant to generative AI, we developed a novel dataset of TV and movie plots with their associated content ratings as a naturalistic measure of content appropriateness. This dataset provides several advantages over traditional content moderation datasets like Jigsaw and TweetEval. While traditional datasets focus on short-form user-generated content from social media and comment sections, our TV and movie plots represent longer-form creative content of the type increasingly produced through generative AI systems.

Using introduced definitions of identity-related speech suppression, we find that across APIs and for essentially all identity groups—excepting some straight and Christian content—identity-related speech is suppressed more than other speech. We additionally find differences between content moderation APIs' behaviors on generative AI content (prompts, responses, and longer media descriptions) versus traditional short-form data (tweets, short comments, etc.). We find that these APIs are more likely to incorrectly flag longer text, but generally perform better on the generative AI content. We also find differences in the type of identity-related speech suppressed between these datasets and in the overall type of content incorrectly flagged, with political content more likely to be incorrectly flagged in the traditional data while TV violence is incorrectly flagged in the generative AI data. We find that identity-related speech suppression is due to stereotypes and text associations specific to each identity. Further study is needed to understand what impact this may have on downstream applications such as the creation of children's books or movie scripts.

Key to our perspective in this work is a focus on the potential of automated content moderation to limit AI-generated or AI-mediated speech. We note that this is different than the traditional content moderation context aimed at filtering speech online generated by real people, in terms of the types of content generated and filtered, its goals, and in that traditional content moderation relies on a mix of automated and human-implemented policies. When considering AI-generated content, any content filters must be real-time, and thus the automation of the content moderation policy is required. In such a context where generative AI is used to create speech or creative content, we argue for a focus on measuring and avoiding incorrect speech suppression, as provided by our introduced measures and benchmark.

Underlying these results are the inherent difficulty of correctly identifying content that is harmful or which a company prefers not to have their generative AI system create. Distinguishing desired from undesired content is a hard problem even in traditional contexts where human judgment is used as part of a partially-automated system. Determining how to practically navigate this inherent tension such that systems are usable while desired speech is allowed is a key problem for the generative AI industry, and one on which we encourage future research.

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